3.1.2 Amount of substance: Sheet 4

Titration calculations · 16 marks · about 25 minutes

Key information

- Mean titre: use concordant titres only (within 0.10 cm^3); never include the rough.
- **n = c × V** with V in dm^3 ($cm^3 \div 1000$).
- Use the equation ratio to go from moles of the solution in the burette to moles in the conical flask.
- Scale up if the sample was taken from a larger volume, eg 25.0 cm^3 from 250 cm^3 → × 10.
- Back titration: moles of excess reagent added – moles left over (from titration) = moles that reacted.

Section A: Self-quiz

01 The table shows some results from a titration.

Titration	Rough	Titre 1	Titre 2	Titre 3
Initial reading / cm^3	0.00	11.30	0.00	8.55
Final reading / cm^3	26.85	37.20	26.20	34.55
Titre volume / cm^3	26.85	25.90	26.20	26.00

What is the correct mean titre?

[1 mark]

A 25.95 cm^3
B 26.03 cm^3
C 26.10 cm^3
D 26.24 cm^3

02 A solution of volume 500 cm^3 contains 150 g of ammonia. What is the concentration, in mol dm^{-3}, of ammonia in this solution?

[1 mark]

A 0.51
B 8.82
C 16.7
D 17.6

03 A student adds 25.0 cm^3 of 2.00 mol dm^{-3} hydrochloric acid to some indigestion tablets.

Calculate the amount, in moles, of hydrochloric acid that was added to the tablets.

Give your answer to an appropriate precision.

[1 mark]

04 A student is asked to prepare a solution of hydrochloric acid with a concentration of 0.0400 mol dm^{-3}. What volume of distilled water should the student add to 25.0 cm^3 of 2.00 mol dm^{-3} hydrochloric acid to prepare this solution?

[1 mark]

A 25 cm^3
B 50 cm^3
C 1225 cm^3
D 1250 cm^3

05 A 25.0 cm^3 sample of diluted sulfuric acid is transferred to a conical flask using a pipette. A few drops of methyl orange indicator are added before titrating the acid with sodium hydroxide.

(Note: Methyl orange is red in acid and yellow in alkali.)

Which colour change is observed at the end point in each titration?

[1 mark]

A Yellow to red
B Red to orange
C Yellow to orange
D Red to yellow

Section B: Exam questions

06 A student dissolves an unknown mass of sodium hydroxide in water to make 200 cm^3 of an aqueous solution.

A 25.0 cm^3 sample of this sodium hydroxide solution is placed in a conical flask and is titrated with 0.150 mol dm^{-3} sulfuric acid.

$2\ NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2\ H_2O(l)$

Table 1

Titration	Rough	1	2	3
Final reading / cm^3	20.75	40.35	21.05	40.60
Initial reading / cm^3	0.00	20.75	1.20	21.05
Titre / cm^3	20.75	19.60	19.85	19.55

Calculate the mass of sodium hydroxide used to make the original solution.

[5 marks]

07 $Mg(OH)_2$ is used as an antacid to treat indigestion.

A student does an experiment to determine the percentage by mass of $Mg(OH)_2$ in an indigestion tablet.

40.0 cm^3 of 0.200 mol dm^{-3} HCl (an excess) is added to 0.200 g of a powdered tablet. The mixture is swirled thoroughly.

All of the $Mg(OH)_2$ reacts with HCl as shown.

$Mg(OH)_2 + 2\ HCl \rightarrow MgCl_2 + 2\ H_2O$

The amount of HCl remaining after this reaction is determined by titration with 0.100 mol dm^{-3} NaOH

29.25 cm^3 of 0.100 mol dm^{-3} NaOH are needed.

Calculate the percentage by mass of $Mg(OH)_2$ in the indigestion tablet.

[6 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**A**	Source: 7404/1 Jun 2024 Q12	1
02	**D**	Source: 7404/1 Jun 2018 Q19	1
03	Moles = $25.0 \times 10^{-3} \times 2.00 = 0.0500$ mol	Source: 7404/1 Jun 2024 Q07.2 (stem shortened) Answer must be to 3 sf	1
04	**C**	Source: 7404/1 Jun 2025 Q21	1
05	**B**	Source: 7404/1 Jun 2018 Q24 (stem shortened)	1
06	M1 Mean titre = (19.60 + 19.55) ÷ 2 = 19.575 cm^3 M2 Moles of $H_2SO_4 = 0.150 \times (19.575 \div 1000) = 2.936 \times 10^{-3}$ mol M3 Moles of NaOH in 25 $cm^3 = 2.936 \times 10^{-3} \times 2 = 5.87 \times 10^{-3}$ mol M4 Moles of NaOH in original 200 $cm^3 = 5.87 \times 10^{-3} \times 8 = 0.04698$ mol M5 Mass of NaOH = 40.0 × 0.04698 = 1.88 g (1.9 g)	Source: 7404/1 Nov 2020 Q02.1 M1 = calculation of mean titre M2 = M1 × 10^{-3} × 0.150; M3 = M2 × 2; M4 = M3 × 8 Allow correct alternative approaches	5
07	M1 Amount of HCl added = 0.200 × 0.040 = 0.00800 mol M2 Amount of NaOH = 0.100 × 0.02925 = 0.002925 mol (= amount of HCl remaining) M3 Amount of HCl reacted with $Mg(OH)_2$ = 0.00800 – 0.002925 = 0.005075 mol M4 Amount of $Mg(OH)_2 = 0.005075 \div 2 = 0.0025375$ mol M5 Mass of $Mg(OH)_2 = 58.3 \times 0.0025375 = 0.148$ g M6 % by mass = 0.148 ÷ 0.200 × 100 = 74.0%	Source: 7404/1 Jun 2023 Q02.6 M3 = M1 – M2; M4 = M3 ÷ 2; M5 = M4 × 58.3; M6 = M5 ÷ 0.200 × 100 Do not allow M6 if > 100%	6